

Mukesh Singh

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RESEARCH INTERESTS

My passion lies in understanding the fundamental properties of nanomaterials, including their electronic, optical, anharmonic, and elastic characteristics, and predicting novel materials using ab initio methods and machine learning approaches. These attributes of material extend my interest in practical applications, such as thermoelectric properties, thermal conductivity, batteries (metal-ion and metal-Sulfur), catalysis reactions like Hydrogen Evolution Reactions and Oxygen Evolution Reactions etc, and hydrogen storage.

RESEARCH EXPERIENCE

Central Michigan University Postdoctoral research fellow First-principles study of electrode materials for metal-sulfur batteries.	Mount Pleasant, USA Jan, 2025 – present
Indian Institute of Technology (IIT) Bombay Postdoctoral research fellow Quantum dot modeling and 2D materials: electronic, optical, and thermoelectric properties.	Mumbai, India Apr 2024 – Dec 2024

EDUCATION

Indian Institute of Technology (IIT) Bombay Ph.D. in Computational Condensed Matter Physics Thesis: Structural and Electronic Properties of 2D Materials and their Applications in Hydrogen Production and Hydrogen Storage: First Principles Study	Mumbai, India 2019 – 2024
Kirori Mal College, University of Delhi M.Sc. in Physics Thesis: Stringy Correction to Reissner-Nordström Black Hole: Multi-Universe Scenario	Delhi, India 2015 – 2017
Deen Dayal Upadhyay Gorakhpur University B.Sc. in Physics, Chemistry, Mathematics	Gorakhpur, India 2012 – 2015

TECHNICAL SKILLS

Projects: Vestapy, RxnPy, VaspWorks, Pre_and_post_processing_dft (see github.com/mukesh4iitb).
Programming: Python, Fortran, Bash (awk, sed), C; C++ (basic), machine learning.
Simulation & DFT: VASP, Quantum ESPRESSO, Gaussian, AFLOW.
Materials Modeling: ASE, Pymatgen.
Analysis Tools: Phonopy, Phono3py, ShengBTE/AlmaBTE, BoltzTrap2.
Software: L^AT_EX, LibreOffice, VESTA, XCrysDen, GIMP, Inkscape, Origin-lab.

PUBLICATIONS

- [1] Jyoti Pandey, Aliakbar Yazdani, **Mukesh Singh**, .., and Bradley D. Fahlman. “Regulating Polysulfide Conversion on Fe–N_x Single-Atom Catalysts Supported on Graphitic Carbon Nitride Revealed by Semi-In Situ XPS of Li-S Batteries”. (Submitted). 2026.
- [2] **Mukesh Singh** and Veronica Barone. “Unified Thermodynamic and Kinetic Criteria for Screening 2D Anchoring Materials in Li–S, Na–S, and K–S Batteries”. (Submitted). 2026.
- [3] **Mukesh Singh**, Surinder Pal Kaur, and Brahmananda Chakraborty. “Modeling and tuning the electronic, mechanical and optical properties of a recently synthesized 2D polyaramid: a first principles study”. In: *Phys. Chem. Chem. Phys.* 26 (July 2024), p. 21874. DOI: 10.1039/D4CP02027H. URL: <https://doi.org/10.1039/D4CP02027H>.
- [4] **Mukesh Singh**, Alok Shukla, and Brahmananda Chakraborty. “Improving hydrogen evolution catalytic activity of 2D carbon allotrope biphenylene with B, N, P doping: Density functional theory investigations”. In: *International Journal of Hydrogen Energy* (2024), pp. 569–579. DOI: <https://doi.org/10.1016/j.ijhydene.2023.08.359>.
- [5] Pratap Mane, Surinder Pal Kaur, **Mukesh Singh**, Ajit Kundu, and Brahmananda Chakraborty. “Superior hydrogen storage capacity of Vanadium decorated biphenylene (Bi+V): A DFT study”. In: *International Journal of Hydrogen Energy* 48 (2023), pp. 28076–28090. DOI: <https://doi.org/10.1016/j.ijhydene.2023.04.033>.
- [6] **Mukesh Singh** and Brahmananda Chakraborty. “2D BN -biphenylene: structure stability and properties tenability from a DFT perspective”. In: *Phys. Chem. Chem. Phys.* 25 (2023), pp. 16018–16029. DOI: 10.1039/D3CP00776F.

- [7] **Mukesh Singh**, Alok Shukla, and Brahmananda Chakraborty. “High capacity hydrogen storage on zirconium decorated γ -graphyne: A systematic first-principles study”. In: *International Journal of Hydrogen Energy* 48 (2023), pp. 37834–37846. DOI: <https://doi.org/10.1016/j.ijhydene.2022.07.062>.
- [8] **Mukesh Singh**, Alok Shukla, and Brahmananda Chakraborty. “Highly efficient hydrogen storage of a Sc decorated biphenylene monolayer near ambient temperature: ab initio simulations”. In: *Sustainable Energy Fuels* 7 (2023), pp. 996–1010. DOI: 10.1039/D2SE01351G.
- [9] **Mukesh Singh**, Alok Shukla, and Brahmananda Chakraborty. “An Ab-initio study of the Y decorated 2D holey graphyne for hydrogen storage application”. In: *Nanotechnology* 33 (July 2022), p. 405406. DOI: 10.1088/1361-6528/ac7cf6. URL: <https://dx.doi.org/10.1088/1361-6528/ac7cf6>.

AWARDS & SERVICE

Awards: CSIR-SRF (2021); CSIR-JRF (2017, 2018); INSPIRE Fellow (Top 1%, 2012); NGPE: Top 1% (2015), Top 10% (2013, 2014).

Positions of Responsibility: Academic Unit Representative (AURAA, 2020–2022); General Secretary, RSAP (2020–2021); System Administrator, H14 IIT Bombay (2022–2023).

Teaching Assistant: Teaching Assistant for courses in Python, Fortran simulations, E&M, and Thermal Physics.

REFERENCES

Prof. Alok Shukla (Ph.D. Supervisor)

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Department of Physics
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Prof. Aftab Alam

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